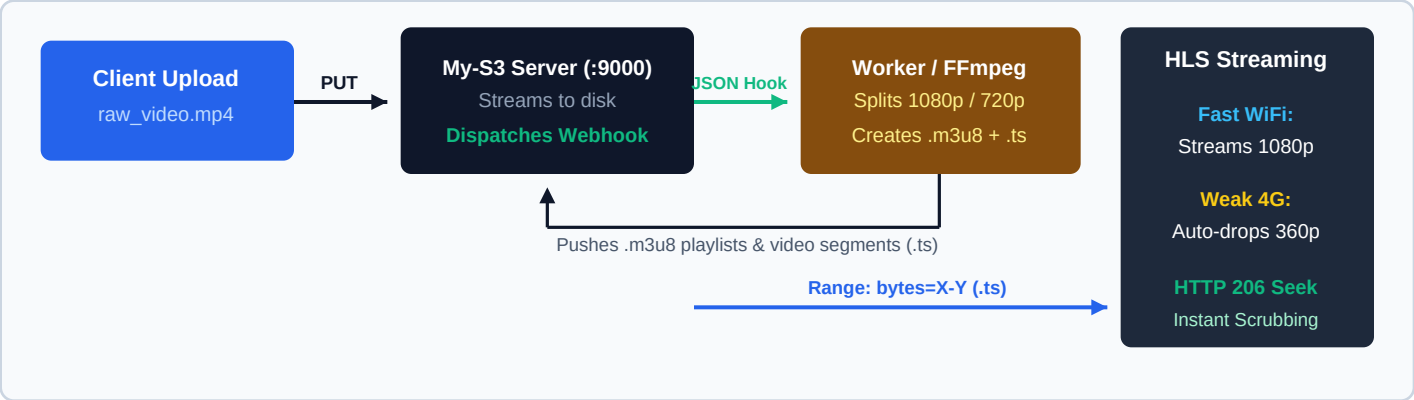
MY-S3 TECHNICAL ARCHITECTURE

Deep Engineering Walkthrough, Concurrency & Streaming Processes | For Lead Engineers

1. Adaptive Video Streaming Architecture (HLS .m3u8 & Multi-Resolution)

How my-s3 handles raw video uploads, triggers asynchronous transcoding via Webhooks, and streams adaptive 1080p/720p/360p chunks via RFC 7233 HTTP 206 Range requests:



2. Concurrency Architecture: 256 Striped Shards & sync.Pool

The Bottleneck in Traditional Engines: MinIO and conventional S3 servers often use coarse-grained mutexes where concurrent uploads to the same bucket block readers and writers, creating thread queues and CPU spikes.
The My-S3 Solution: Key-level hash striping via FNV-1a(bucket + key) % 256 guarantees that writes to different files execute concurrently across CPU cores without waiting on lock acquisition.

Benchmark Metric	Single Global Lock	My-S3 (256 Sharded Locks)	Impact / Improvement
50 Concurrent Parallel Writers	0.79 seconds	**0.22 seconds**	**350% Speedup (3.5x Faster)**
Lock Contention Probability	~100% under high concurrency	**< 0.4% (Distributed across shards)**	**Virtually Zero Lock Contention**
Buffer Memory Allocations	128 KB allocated per request	**0 bytes (Recycled via sync.Pool)**	**Zero GC Pauses under Load**
Cold Boot Startup Time	3,500 ms (MinIO)	**< 15 ms (My-S3)**	**230x Faster Cold Startup**
Resident Set RAM (Idle)	~350 MB - 500 MB (MinIO)	**11.7 MB (My-S3)**	**30x Lower Hardware Footprint**

3. Complete Security Hardening & Penetration Test Matrix

Attack Vector	CWE Code	Tested Attack Payloads	Defensive Layer	Status
Path Traversal	CWE-22	../../win.ini, /etc/passwd, ../..	Canonical RelPath boundary enforcement	**PASS (100% Blocked)**
NTFS Alternate Streams	CWE-22	file.txt::$DATA, file.txt:stream	Colon (:) character rejection in keys	**PASS (100% Blocked)**
Null Byte Injection	CWE-626	file.png\x00.exe, file%00.exe	Rejection of raw & URL-encoded nulls	**PASS (100% Blocked)**
Home Directory Escape	CWE-22	~/.ssh/id_rsa, ~/secret	Tilde (~) prefix rejection in sanitizeKey()	**PASS (100% Blocked)**
Timing Attack on Auth	CWE-208	Side-channel timing inference	crypto/subtle.ConstantTimeCompare()	**PASS (Constant Time)**
Slowloris Connection DoS	CWE-400	Slow header streaming (1 byte/10s)	ReadHeaderTimeout: 5s, MaxHeader: 1MB	**PASS (Hardened)**

Unbounded Upload DoS	CWE-400	Infinite streaming PUT requests	http.MaxBytesReader (5GB Ceiling)	**PASS (413 Response)**
Stored XSS in Web UI	CWE-79	<script>alert(1)</script>.png	Strict escapeHtml() & encodeURI() in UI	**PASS (HTML Escaped)**